

AI Career Intelligence Platform

DevOps Design Document

Infrastructure • Docker • Kubernetes • CI/CD

Version: 1.0

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1. Infrastructure

The platform follows a cloud-ready architecture with separate frontend, backend, database, cache, and AI services.

Component	Technology
Frontend	Next.js
Backend	FastAPI
Database	PostgreSQL
Cache	Redis
AI Services	LangChain, spaCy, scikit-learn
Reverse Proxy	Nginx

2. Docker

Each application component runs in its own Docker container for portability and consistent deployments.

- Separate containers for frontend, backend, PostgreSQL and Redis.
- Use Docker Compose for local development.
- Environment variables managed using .env files.
- Persistent volumes for database storage.

3. Kubernetes

Kubernetes provides orchestration, scaling and high availability for production deployments.

Resource	Purpose
Deployment	Manage application replicas
Service	Internal networking
Ingress	External access
ConfigMap	Configuration
Secret	Sensitive credentials
Horizontal Pod Autoscaler	Automatic scaling

4. CI/CD Pipeline

Recommended pipeline:

Developer → GitHub → GitHub Actions → Build → Test → Docker Image → Container Registry → Kubernetes Deployment

- Run linting and unit tests automatically.
- Build Docker images for every release.
- Deploy to staging before production.
- Monitor deployment health and support rollback.

5. Conclusion

This document defines the DevOps strategy for Version 1.0 of the AI Career Intelligence Platform, covering infrastructure, containerization, orchestration and automated delivery.